Ke-Chi (Arc) CHANG

Taipei/Taiwan | +886-934-335-141

sky1236@gapp.nthu.edu.tw

https://arcchang1236.github.io/

https://www.linkedin.com/in/arcchang1236/

WORK EXPERIENCE & EDUCATION

Mediatek Inc. *2020 - Now*

Senior Software & AI Engineer

- Developed *more than10 AI algorithms* for Mobile/Tablet/Surveillance Cameras.
  - Low-level vision (Denoising, Deblurring, Super Resolution, HDR)
  - High-level vision (Detection, Semantic/Instance Segmentation, Facial Landmark Detection)
  - Other ISP-related tasks (Demosaicing, Remosaicing, Noise Modeling)
- Integrated the *software C-Model* between chip hardware and algorithms.
- Led and contributed to *more than 10 projects* for different clients. (OPPO/Vivo/Xiaomi/Honor)

National Tsing-Hua University (NTHU) *2018 - 2020*

M.S. - Computer Science [Thesis Advisor: Hwann-Tzong Chen]

- Conducted 5 industry-academia collaborative projects and published 1 paper in a top-tier conference.

National Tsing-Hua University (NTHU) *2014 - 2018*

B.S. - Computer Science

- Graduation project research is *"360° Video Object Detection"*, awarded 3^{rd} place.)

PUBLICATIONS

Learning Camera-Aware Noise Models

Europeon Conference on Computer Vision (ECCV) 2020

- Proposed a generative model CA-NoiseGAN for estimating real-world noise.
- Outperforming state-of-the-art noise models by more than 80% in KL divergence.
- Training the denoising network with generated images by our noise model can improve PSNR by about 1dB.

OTHER WORK EXPERIENCES

Mediatek Internship *2019.07 - 2020.06*

Study and develop learning-based methods of RAW denoising tasks.

PEGATRON Internship *2015.07 - 2015.09*

[Internship] Develop Android APP of access control system with face recognition.

Umbo Computer Vision *2018.11 - 2019.08*

[Industry-Academy Cooperation] Multi-person 3D pose estimation by learning-based methods.

Automotive Research Testing Center (ARTC) *2018.03 - 2018.11*

[Industry-Academy Cooperation] Self-driving location technology by SLAM (error in 1m).

Industrial Technology Research Institute (ITRI) *2017.09 - 2018.08*

[Industry-Academy Cooperation] Use learning-based method to improve the sky noise of SLAM.

Conference on Computer Vision, Graphics, and Image Processing (CVGIP 2018) *2018.08*

2^{nd} place in *"Indoor 3D deep space construction competition based on deep learning and 2D images"*.